HANDOVER / 01

CS-80 Reference

DRAFT / Source qualification current; merge and release evidence pending

cheshmakzanoon-ops/yamaha-cs80 | C++20 / JUCE 8.0.11 | AGPLv3

This is an interim engineering handover for PR #6, not a completed release. CPU budgets and original physical-PANEL musical click scores are still outside their unchanged targets. The three-rate final-audio engine audit is currently running; no unobserved pass is claimed.

The proven Linux validator-host lifetime failure is repaired, the nodal top-octave numerical law and all eight host-feature families are integrated, and all 25 disputed tone cells carry schematic-page hypotheses. No original instrument was measured.

Recorded source and landing

Record	Exact identity / result
Production audio	83b7f7670f000a95c6b44f96ffb0df3894755d14
Circuit V2 baseline	83b7f7670f000a95c6b44f96ffb0df3894755d14
Main merge	PENDING_MAIN_MERGE
Landing PR	Pull request and review history
Release source	PENDING_RELEASE_SOURCE
Release result	PENDING: feature branch is not merged; no new release run Installer workflow and artifacts

Required workflow evidence

Workflow	Result	Tested source
Main build + pluginval	PENDING	PENDING_MAIN_MERGE
Main DSP measurements	PENDING	PENDING_MAIN_MERGE
Main screenshots	PENDING	PENDING_MAIN_MERGE

Read docs/STATUS.md first for the current model register and open issues. This document is a dated snapshot. A successful validator or software reference gate does not establish a hardware match or replace the owner's real-DAW sessions.

WORKING DRAFT - RESOLVE PENDING FIELDS

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Scorecard and measurement scope

Before and after values retain their source, rate and measurement context. All acceptance limits stay unchanged.

Metric / rate	Before / baseline	Qualified after	Target	State
Eight held voices; 48k/256	main .588123-.696543	final83 .605709-.888572	.15-.20 RTF	OPEN
Quiet idle; 48k/256	main .143823	final83 .067570 (6.76%)	<=.020 (2%)	OPEN
Full-bank peak	Preplan +3.9 dBFS	Final83 full audit PENDING	<=0 dBFS	UNVERIFIED
Voice-steal click	Preplan 1.17	Final83 full audit PENDING	<=1	UNVERIFIED
Ring-off alias; 48k	Preplan -33.8 dB	Final83 full audit PENDING	<=-70 dB	UNVERIFIED
PANEL Bass retrigger	Earlier 1.51 at 44k	Pre-final .568/1.441/1.365	<=1 at all rates	OPEN
Funky note-off	Earlier up to 1.65	Pre-final 1.627/1.291/1.221	<=1 at all rates	OPEN
Nodal 2x/4x top band	Original top band fails	.04884/.04392/.002967/.001954/.0000583	Original rate limits	PASS
ADJ p11 HP / LP	4.3+/-1 / 2.2+/-1 Vpp	Model 1.692 / 1.648 Vpp	HP 3.3-5.3; LP 1.2-3.2	HP OPEN
Legacy / V1 / V2 audio	Immutable baselines	All four byte-identical	Exact bytes	PASS

CPU: [run 41292549292] twelve immutable sources, 36 serial rounds, AMD EPYC 7763, 48 kHz/256, 1,024 callbacks. Main-to-final idle improves 53.02%; held p99 reaches 1.050427 RTF with 49 selected held misses. CPU percentages express callback realtime factor, not OS load.

The Bass/Funky figures shown are pre-final b75 physical-PANEL/tone-tab causal observations, not the final83 full-bank audit. Bass excess peaks +7.8 ms after onset; Funky omitted-note-off control still exceeds one. Neither is a proven charge discontinuity.

Historical 00ceccf peak/steal/alias figures remain historical. The running all-suite audit must supply final83 results. Nodal voltage predictions and native M41/tone images are source-based model evidence, not analogue measurements.

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The six completion workstreams

Implementation status and acceptance status are recorded separately.

1. CPU

IMPLEMENTED; BUDGETS OPEN

Oscillator event and Jacobian caches, sixteen component chains, split process and exact idle/FIR savings. Final held .605709-.888572; idle 6.757%. True independent-rank SIMD remains open.

Published changes: 6bf6055, 031a333, 1207068, 23a7b8d, ee4377f

2. Retrigger / note-off

STATE DEFECTS FIXED; MUSICAL SCORES OPEN

Filter charge/time, ring phase and connected VCA coupling corrected without fades. Physical PANEL Bass 48/96k and Funky note-off remain above score 1; final causal audit pending.

Published changes: 481c7e9, b75b4d6, 83b7f76

3. Linux pluginval 139

PROVEN HOST MECHANISM REMOVED

Original/fixed/restored JUCE callback maps prove use-after-free, pass, use-after-free. CI pins SHA-checked pluginval2 beta3; strict10, headless/GUI/fixed-seed, one attempt.

Published changes: f39af47; VALIDATOR_HOST_LIFETIME.md

4. Nodal filter

TOP-OCTAVE PASS; HP SETUP OPEN

Original 2x/4x error limits pass, including 176.4k. M41 p41: 150k series CP7-AI and distinct 33k shunt. ADJ printed p11 HP model 1.692 Vpp vs 4.3+/-1; no fit.

Published changes: 4917708, 99d5198; M41/ADJ native evidence

5. Host/UI and modern

IMPLEMENTED; DAW SESSIONS OWNER-ONLY

18 timed automation lanes; actual 0BSD MTS client; circuit memory audio; both-on/off chorus/tremolo; switch/miniature MIDI learn; undo host gestures; engine taps and 16-voice ownership. Neutral in PURE.

Published changes: 6474265, fa2016c2, 6f1c4aa, 6300163

6. Tone cells

ALL 25 TRANSCRIBED; HYPOTHESES REMAIN

Native Yamaha source pages S31-S38: T51 eight, T52 four, T53 nine, T54 four. Glyph/board-revision uncertainty stays labelled; no V1 numeric or fingerprint modification.

Published changes: 5a3027d, c56070f

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Evidence and compatibility

The immutable audio boundaries and the causal proof remain part of the deliverable.

Capture	Immutable baseline / current V2 pin	Result
Legacy default	d56911a7c5f6953cd244457226b126cc567a538b	Byte-identical
Legacy tails	d56911a7c5f6953cd244457226b126cc567a538b	Byte-identical
Circuit V1 numerics	22d6a32c48c85f9b813506edcd856f73f4293c2d	Byte-identical
Circuit V2 audio/state	83b7f7670f000a95c6b44f96ffb0df3894755d14	Byte-identical; strict

Intentional circuit-audio chronology

Retained ring phase / precise charge

Audio: 481c7e9fdb20372288864eecf746bf7efb28190a
Separate V2 pin: 8a85faab1a125cc14425d9184f7022f6746e40a6
Strict compatibility / status evidence: Predeclared event gates + strict STATUS

Connected idle VCA coupling

Audio: b75b4d653b6ed6f0aa5ef8ee99b670d83e975250
Separate V2 pin: fb43c04e68e91eea120add3a89e6866ef1e64e01
Strict compatibility / status evidence: VCA state/causal mutation gates, strict STATUS

Current-sample tail-pan ownership

Audio: 83b7f7670f000a95c6b44f96ffb0df3894755d14
Separate V2 pin: 14ef653ed60be67ff98656966f9181c5e9b6b4fc
Strict compatibility / status evidence: Tail-pan modulation 618+6 assertions, strict STATUS

Why the evidence supports the claims

The 36-round final CPU archive independently verified 346 artifact hashes, 884 source hashes, 1,836 finite rows and twelve source summaries. GitHub Actions run 41292549292; SHA-256 3ffa33d3d5674ea0f1aed409efb3d8940024a8891f4e51147a1d69b3f37dffed.

M41 service p41 and ADJ printed p11 are documentary references. The corrected LP 150k series leg and separate 33k shunt do not resolve the HP 1.692 Vpp discrepancy.

All 25 tone cells cite native S31-S38 board and schematic sheets. Ambiguous T53 B14 installed marking versus schematic value remains a revision hypothesis; V1 fingerprint e5f877a025fc52c6 is frozen.

Future changes: commit tolerances before evaluating; mutation-check every new gate. Push a circuit-audio commit, make its separate full-SHA V2 pin commit, then run scripts/check-compat-gates.sh --strict-v2 and record the result in STATUS. Never weaken, skip or delete a gate to obtain a pass.

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Remaining work and owner boundaries

An unresolved target stays open, with the evidence needed for the next decision.

Held and idle CPU speed

Need measured 3.03-4.44x held and 3.38x idle acceleration. Preregister truly masked 16-rank SIMD, exact states/rounding/NaN oracle, mutation controls and same-host three-before/after benchmark. Event cache alone is insufficient.

PANEL Bass and Funky click scores

Original <=1 gates remain red at named rates. Compare time-varying attack to high-precision nonlinear reference and use diagnostic-only pedestal ablation before proposing any physical law change; no fade.

ADJ HP setup loading

Verify source and mixer impedances, VR7-VR9 alignment and available manual wiring for service p11 before altering the model; never fit to one 4.3+/-1 Vpp point.

Native tone historical applicability

Inspect board revision/installation evidence for all 25 hypotheses; retain S31-S38 readings, ambiguity labels and immutable V1 numerics until corroborated.

PR6 main acceptance and release

PENDING complete all-rate raw audit and final main CI. Then obtain accepted merge M, verified main screenshots S, guarded release and true installer URLs; replace this draft PDF with six-page final.

Owner-only actions - outside this task

- Apple/Windows signing, notarization credentials and certificates
- Real DAW session checks on the requested installed hosts
- Legal trademark/nameplate review of Yamaha and CS-80
- AAX/Avid/PACE integration

Evidence rule: cite the schematic/manual page for each physical law, or label it a declared inference and retain its alternatives. The HP setup discrepancy and ambiguous tone-cell revisions do not justify curve fitting. No original instrument was measured in this work.

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Reproduce and continue

Use the exact source identities above. Run the required checks after each relevant change.

```
cmake -S . -B build-core -G Ninja -DCMAKE_BUILD_TYPE=Release -DCS80_BUILD_PLUGIN=OFF \
  -DCS80_BUILD_TESTS=ON

cmake --build build-core && ctest --test-dir build-core --output-on-failure

scripts/check-compat-gates.sh --strict-v2

cmake --preset linux-release && cmake --build --preset linux-release && ctest --preset linux-release \
  --output-on-failure

build-core/cs80_engine_audit --suite all --rates 44100,48000,96000 --out artifacts/audit

build-core/cs80_source_factory_benchmark --rate 48000 --block 256
```

UI smoke requires Xvfb, xauth and openbox. An isolated Openbox exit 70 is an environmental display failure; retain its log and rerun the unchanged check. CPU comparisons require at least three rounds per source and serial measurement after builds finish.

Primary references and retained evidence

Current STATUS / scorecard

Final 36-round CPU qualification

Validator callback lifetime

Native M41 / ADJ evidence

25 native tone-cell readings

Continuation rules

Keep parameter IDs and ordering append-only, use enum names, and select the next unused AU version hint (16 or higher). Keep modern additions neutral by default and inert in isPureCs80(policy). Before copying or replacing APVTS state, take getUndoableState().treeLock() and call syncNow(). Use small reviewable commits; never force-push or rewrite published history.

WORKING DRAFT - RESOLVE PENDING FIELDS